Public-Safety Answering and Dispatch Facility-Class Design Basis

The First Facility Class of the Emergency Services Sector Design Basis

DBA-ESS-AD-FC1 — Facility-Class Design Basis, DBA-ESS Series

Issued Under: Emergency Services Sector Framework (DBA-ESS_Sector_Framework v0.7); DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for public-safety answering and dispatch: the 911 centers that answer emergency calls, the systems that tell them where the caller is and which units to send, and the radio systems that carry their instructions to the field. It is the first facility class in the DBA-ESS series. The *Emergency Services Sector Framework* (DBA-ESS_Sector_Framework) states the sector's design basis in general terms, and this document applies it to answering and dispatch. It names the events a 911 center must keep working through, traces how each reaches the people who need help, sets the measures that size its protection, and states who must authorize what. It describes no particular center or jurisdiction. A design basis for a specific center or answering area starts from this document and adds only what is particular to it.

A 911 center is the point at which the public asks for help and a response begins. When it cannot answer, or can answer but cannot send anyone, every other failure in its area grows worse, because the response that would have contained that failure does not come. The framework calls this sector the operate-through sector (framework §1.5). This class is where that requirement is sharpest: a 911 center is busiest at exactly the moment its power, its networks and its neighbors are failing.

1.2 Scope

The class covers the facilities that answer emergency calls and direct the response: the public safety answering points, which this document calls 911 centers, and their backup centers; their buildings, power plants and the rooms in which telecommunicators work; the call-handling equipment that presents calls and the caller's location; computer-aided dispatch, the software that records each incident, shows where every unit is, recommends which units to send and tracks them to the end of the call; the Emergency Services IP Network of a Next-Generation 911 system, with the core services that route calls across it; and the agency-owned public-safety radio systems, chiefly P25, with their consoles, sites and controllers. It covers the systems that monitor and manage these facilities, and the telecommunicators, supervisors and technicians who run them.

The class stops at four boundaries. On the author's decisions of 6 October 2026, the boundary with the communications sector lies where the call and the caller's location enter the 911 system. Communications carries them that far: the originating networks and, in an enhanced 911 system, the 911 trunks, selective routing and location delivery that a carrier or a 911 service provider operates. From that point inward the facilities are this class's. The Emergency Services IP Network and agency-owned P25 systems are this class's by ownership and consequence. FirstNet, a carrier network built for public safety, belongs to DBA-COMM-AC-FC2 (*Access and Edge*), with an elevation obligation. Response units, their stations and the radios they carry

belong to DBA-ESS-RF-FC2. Emergency operations centers and the origination of public alerts belong to the coordination class, DBA-ESS-EMCo. The boundaries are developed in Section 8.3.

1.3 The Reference Configurations

Three configurations are used as reference cases.

A county 911 center with one backup center is the governing configuration. It answers every call for its area, dispatches police, fire and emergency medical units, and runs its own computer-aided dispatch and radio consoles. Most of the country is served by centers of this kind, and it sets the class's first measure.

A consolidated regional center on a Next-Generation 911 network shows the second configuration. It serves several jurisdictions, owns or shares the network and core services that route calls to it, and can hand calls to other centers on the same network. Its loss reaches farthest, and its network makes transfer possible on a scale a legacy center cannot match.

A small center with a single position, in a rural county, shows the third. It has one or two telecommunicators on shift, no backup center of its own, and depends on a neighbor to take its calls when it cannot.

A requirement that names one configuration applies to that configuration. A requirement that names none applies to all of them.

1.4 Class Type

This is a physical class, on the author's decision of 6 October 2026, under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it can do comes mainly from the loss of a place and the things in it: a center, its backup, its power, its consoles and its radio sites. Computer-aided dispatch and the Next-Generation 911 network have logical features, and Sections 4.4 and 4.7 treat them, but they matter here because their failure can take a center's ability to answer or dispatch out of service at once.

1.5 Relationship to Governing Documents

This instrument is issued under the Emergency Services Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, principles or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 10 what the framework should take up. References of the form "framework §N" are to the Emergency Services Sector Framework in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that

obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, including in advance, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*The Envelope*) reserves the term "envelope" for the pre-authorized envelope within which advance discharge is valid; this instrument uses the word in that sense only, and calls a region of normal operation whose limits are monitored an operating band. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

The communications design basis is cited, not inherited. DBA-MA-CDB (*Military Action Communications Design Basis*) sets out seven requirements, CDB-1 through CDB-7, for every facility in every sector whose communications may be degraded or denied. For this class the degraded condition those requirements describe is not an excursion but the ordinary operating environment (framework §1.5), and Section 7 states how the seven apply, as *Military Action Communications Design Basis* §3 requires.

2. The Public-Safety Answering and Dispatch Facility Class

2.1 Defining Characteristics

Three features define the class.

First, it must work when everything around it is failing. A 911 center is dispatched into the conditions other facilities are designed to survive: a grid outage, a storm, a network failure, an attack. Its call volume rises as its supports fall away. Its design basis is therefore stated as an operate-through requirement — how long it keeps answering and dispatching through a defined degradation — rather than as a withstand requirement (framework §1.5, §3.5).

Second, its consequence is out of all proportion to its size. A center may be a single room with a few positions, yet it is the only place in its area where an emergency call is answered. The loss of that room is the loss of the public's ability to summon help, whatever the state of the response forces it directs.

Third, it is the junction of networks it does not own. Calls and their location data arrive over carrier networks; radio traffic leaves over the center's own system; computer-aided dispatch increasingly runs in hosted facilities; and the center coordinates with neighbors, hospitals and emergency managers over whatever links survive. The class is defined by what it must do when several of those links have failed at once.

2.2 Deployment Typologies and Size Range

The class spans a wide range. At one end is a rural center with a single telecommunicator at night, no backup of its own and a written agreement with a neighbor. At the other is a regional center with dozens of positions, its own Next-Generation 911 network and core services, a hosted dispatch system, and a radio system covering several counties. Between them are the city and county centers, many of them separate for police and fire, that answer most of the nation's calls. The class design basis applies to all of them, and the values of its measures differ widely across them.

2.3 Consequence Context

The class's governing harm is the loss of the ability to answer emergency calls in an area. Its second harm is the loss of the ability to direct a response once a call is answered. Its third is the loss of the radio link that carries instructions to the units in the field.

Five events are the reference events for the class. The framework confers them at its §9.1, as DBA-ESS-RE-1 through DBA-ESS-RE-5, under the author's decision of 6 October 2026 that every reference event a class proposes is conferred. They are listed here in the order the framework numbers them.

The derecho of 29 June 2012 is the power case. A fast-moving windstorm crossed the Midwest and Mid-Atlantic. The Federal Communications Commission found that seventy-seven 911 centers in six states, serving more than 3.6 million people, lost some service, and that seventeen in three states, serving more than two million, lost it entirely; Fairfax County, Virginia, with about 1.1 million residents, completed no 911 call for seven hours and was not fully restored for three days. Most of the causes the Commission named were on the communications side of the boundary — backup generators that failed in carrier central offices, 911 circuits without physical diversity, and monitoring that lost power with them — and the consequence fell on 911 centers that were themselves powered and staffed. In West Virginia eight centers lost service through failures at the center, among them generators that did not start (DBA-ESS-RE-1; *Derecho 911 Outages* F-6).

The multistate 911 outage of 9 and 10 April 2014 is the transfer case. A software counter in a 911 service provider's call-routing center in Englewood, Colorado, reached its limit and stopped assigning calls to 911 trunks. For up to six hours, eighty-one 911 centers in seven states received no calls through that route, more than 6,600 calls failed, and more than 11 million people were potentially unable to reach help; every county in Washington State was affected. A redundant routing center in Miami was available throughout, and service returned almost at once when calls were switched to it. They were not switched for about six hours, because the failure was not recognized (DBA-ESS-RE-2).

The telephony denial-of-service attack of October 2016 is the volume case. A link posted to social media carried code that made iPhones dial 911 repeatedly. It was opened nearly two thousand times. The 911 center in Surprise, Arizona, received more than a hundred calls within

minutes and was in danger of losing service, and centers elsewhere in Arizona and in California and Texas were affected. A 911 center cannot refuse a call, so its capacity can be consumed by calls that are not emergencies (DBA-ESS-RE-3). The Sheriff's Office's figure of nearly two thousand opens and the later reported figure of 117,502 are not reconciled in the record, and both are stated with their sources (*Call Flood* F-5).

The ransomware attack on Baltimore's computer-aided dispatch on 25 and 26 March 2018 is the first dispatch case. A change to a firewall made while troubleshooting left the system exposed, ransomware reached it, and computer-aided dispatch was out of service for about seventeen hours. Calls were answered, and units were dispatched by voice and by hand (DBA-ESS-RE-4). The seventeen hours is the city's figure; the city's published dispatch records show a gap of about twenty-two. The city said the attack affected messaging functions within computer-aided dispatch (*Baltimore Dispatch* F-6).

The ransomware attack on the City of Dallas, beginning on 3 May 2023, is the second dispatch case. The Royal ransomware group took down city systems including the computer-aided dispatch used by police and fire. Telecommunicators wrote out assignments by hand and dispatched by radio while the system was restored, and 911 calls continued to be answered (DBA-ESS-RE-5). The attackers had been inside the city's network since about 7 April, using an ordinary service account. The units' mobile data and the fire stations' alerting were lost with dispatch, which returned to full automation on 11 May. The city said disruption was minimal; the police and fire unions described missed calls and units sent to wrong locations; no measure was published (*Dallas Ransomware* F-7).

Hurricane Katrina in 2005 is cited as an illustration and is not conferred: more than thirty-five 911 centers were put out of service by the count of the communications panel convened after the storm, and thirty-eight by the House select committee's, and the loss of the center building itself, rather than of its supports, became a design question. Two events already conferred in the communications series also reached 911 service and are cited rather than conferred again: the Nashville bombing of December 2020, DBA-COMM-RE-4, and the AT&T outage of February 2024, DBA-COMM-RE-7.

The design-basis reading of the five is direct. In the first two, the 911 centers were intact and staffed and the calls did not arrive; what decided the consequence was how long it took to get calls to a center that could answer them. In the third, calls arrived faster than any center could answer them. In the last two, calls were answered and dispatch went on without its software, because the people in the room could do it by hand. Those are the class's measures. All five instruments were written on 7 October 2026, and this revision takes in what they returned.

3. The Facility-Class Design Basis

3.1 Why Answering and Dispatch Is the First Class

Answering and dispatch is developed first because it is the sector's tightest coupling with the communications sector and the operational counterpart of that sector's critical-services overlay (framework §3.2, §6.1). Where the boundary between the two falls decides what the overlay must still govern, and this class draws it (Section 10.3). It is also the function on which the other two classes depend: response forces cannot be sent, and emergency management cannot coordinate them, if the call is never answered.

3.2 The Two Governing Measures

Two measures govern the class, on the author's decision of 6 October 2026.

Operate-through duration is how long a 911 center keeps answering calls and directing a response, without outside support, through a defined compound degradation: loss of commercial power, loss of its primary communications links, loss of computer-aided dispatch, and loss of routine road access for staff and fuel (framework §3.5). It is counted for each center and set against the duration of the compound event its area must be designed for. Endurance that depends on fuel deliveries or relief staff counts only what is on hand unless the deliveries or relief are themselves protected and assured. Backup power counts toward it only when it has been shown to start and carry its load (Section 4.1).

Alternate-answering transfer time is how long it takes, after a center loses the ability to answer, until calls for its area are ringing at a staffed position somewhere else — a backup center, a neighboring center or a deployable position — with the caller's location delivered. It is counted from the caller's side. It does not matter to the caller whether the failure was in the center, in the network that routes calls to it, or in a service provider's routing center: the clock runs until the call is answered. The 2014 outage shows why the measure is drawn this way. A working alternate existed for the whole six hours, and the transfer time was six hours because no one recognized that calls had stopped. The 2014 outage also shows what the measure contains. Invoking the alternate took moments once it was begun, and every minute of the six hours was spent recognizing the failure. A transfer time is therefore bounded by recognition time, and each area sets its recognition time as part of its transfer time: a transfer that takes seconds once begun is worth nothing if beginning it waits on a recognition that takes hours (*Multistate 911 Outage* F-1).

The design basis requires that, for the compound event an area must withstand, every 911 center keeps answering and dispatching for the design duration, and that if a center is lost, calls for its area reach a staffed alternate within the transfer time set for the area.

On the author's decision of 11 October 2026, the framework states the form of the operate-through metric for the sector (*Emergency Services Sector Framework* §3.5), and this class applies it. A center's operate-through duration is one link in the sector's chain from call to

response, and the sector's duration is the least of the links, so a center's endurance beyond the units' buys nothing until theirs is raised (DBA-ESS-UC-1). The fuel and staff a center counts on hand are counted once across the sector, in the sector's record, where the radio sites, the stations and the apparatus draw on the same stock.

3.3 The Three Consequence Chains

Chain A — answering. Calls do not reach a staffed position with the caller's location. The cause may be upstream, in the networks that deliver calls; in the center, through loss of power, building or call-handling equipment; or in volume, when calls arrive faster than they can be answered. Its size is set by the population of the answering area and the time until calls are answered again, which is the transfer time.

Chain D — dispatch. Calls are answered, but the center cannot direct a response well: computer-aided dispatch is lost, unit locations are unknown, or the recommendations the center relies on are wrong. Its size is the delay and misdirection of responses across the area, and it is held down by the center's ability to dispatch by hand.

Chain F — field. The center cannot reach its units because the radio system or its links have failed. Its size is the number of units and the area that fall out of contact, and the time until a fallback channel carries traffic.

3.4 Design Parameters at Class Resolution

The two measures are the class parameters, together with a third: the alternate-routing map, the record of where each answering area's calls go when its center cannot take them, which center takes them, whether that center receives the caller's location and can see the area's units, and what must happen for the transfer to begin. A center that cannot produce this map cannot show what its transfer time is. The values depend on the area and are not given here as numbers (Section 10.1). The map also records what invokes each transfer and how long that takes; what each alternate shares with its primary, in design, software and configuration as well as in physical elements; and, for each element upstream of the centers, the population whose calls pass through it. In 2014 one routing component served more than eleven million people in seven states, the alternate could be invoked only by a person who had recognized the failure, and it carried a counter of the same kind as the primary (*Multistate 911 Outage* F-4, F-5, F-6). On the author's decision of 7 October 2026, concentration is recorded in the map rather than made a fourth parameter, and the safeguards on an upstream element — detection, automatic failover and design diversity — are sized by the population it serves, as the coordination class sizes its safeguards by authority reach. A concentration that is not recorded cannot be justified under Section 5.8. The map also holds the area's own evidence, on a cadence the 911 authority sets, of the backup power, the circuit diversity and the monitoring of each upstream element whose loss would stop its calls, and does not rest on what a provider certifies: the federal certification that followed the derecho became a one-time filing in 2026 (*Derecho 911 Outages* F-5). The shared elements it records are verified against the physical route of each circuit, including the links that

deliver the caller's location, at a set interval and on any change; in the derecho Arlington's three location links ran through one office (*Derecho 911 Outages* F-3). And it names overflow alternates whose populations are unlikely to share the audience of an induced flood (Section 5.4).

4. Design Basis Events: Non-Adversarial Set

The framework lists candidate bounding events for the sector but does not develop them (framework §4.1). This section develops the events for answering and dispatch under the series' four headings: the event, the harm that makes it count, what must be done, and for how long. Section 5 applies the military-action extension to them.

4.1 Loss of Commercial Power Outlasting Backup

Postulated event a wide-area loss of grid power lasting longer than the center's batteries and fuel, with resupply delayed by road closures, fuel shortage or competing demand, and with staff unable to reach the center or to be relieved.
Consequence threshold a center that stops answering or dispatching before the design duration of the compound event, or whose radio sites go dark before it does.
Required function every center, backup center and radio site must state its operate-through duration. Each must carry backup power — uninterruptible power for its consoles and call-handling equipment, a generator, and fuel on site — at least equal to the area's design duration, and the generator and its fuel must be protected against the same event that takes the grid. The center must hold a fueling and staffing plan that does not depend on the communications the outage has taken away, including rest and food for telecommunicators who cannot be relieved. Backup power is credited toward operate-through duration only when it has been shown to start automatically and to carry its full load for a stated period, at a set interval and by a test that would reveal the failures of the derecho; a generator not so shown counts as none. In 2012 a carrier office's generators had not been load-tested for at least a year, and generators at several West Virginia centers did not start (*Derecho 911 Outages* F-2).
Design-basis duration the area's design duration for the compound event, which for a major storm or a deliberate attack on the grid is days to weeks.

4.2 Failure of Call Delivery Upstream of the Center

Postulated event a failure in the networks that deliver calls and location to the center — a carrier central office, the 911 trunks, a selective router, a 911 service provider's routing center, or the core services of a Next-Generation 911 network — while the center itself is intact.
Consequence threshold calls for an area not reaching any staffed position, or reaching one without the caller's location.
Required function the center must be able to recognize, from its own call records, that calls have stopped or fallen below the expected rate, and must not depend on the failed network to tell it so. That is necessary and not sufficient: at night the expected rate is low, and in 2014 some centers never noticed at all. Whoever operates an element in the call path must therefore also detect the failure of call completion through it, at a severity set by the consequence — the population whose calls are failing — rather than by the component's view of its own state; in 2014 the failing component issued thousands of alarms and every one was rated minor (*Multistate 911 Outage* F-3). Every finite resource in the call path reports its margin, and its

exhaustion counts as a failure of call completion (*Multistate 911 Outage* F-8). Each area must have an alternate routing path that does not share the failed element, including its design, software and configuration (*Multistate 911 Outage* F-5), and for a failure upstream of the center the transfer to it is automatic on a measured trigger such as the completion rate, rather than waiting on a person's recognition (*Multistate 911 Outage* F-4; Section 6.4). Where the failure lies in a network the center does not own, the center must hold the contacts, the authority and a tested procedure for having calls rerouted. The public must be told by a channel that does not depend on 911 how to reach help in the meantime. The detection and notification placed on the operator of an element are powered and connected independently of that element for the area's design duration, and notification reaches a center at once, not by e-mail to designated staff; in the derecho the carrier's monitoring of its Northern Virginia network lasted thirty minutes after the power failed, and the carrier learned of Fairfax's outage from the county (*Derecho 911 Outages* F-1). The center plans for the states between silence and full service — calls without location, calls on administrative lines, calls rerouted to a neighbor without the area's unit picture — with call-back and verbal location among its procedures, and verifies that any number it tells the public to call instead of 911 does not run through the element that has failed (*Derecho 911 Outages* F-4).
Design-basis duration the transfer time, and beyond it the time to restore the failed element.

4.3 Call Volume Beyond Capacity

Postulated event calls arriving faster than the center can answer them, whether from a mass-casualty event, a regional disaster, or calls generated deliberately or by faulty devices. Two further sources are named. A false or wrong public alert drives a surge: Hawaii's false missile alert multiplied the calls to 911 more than twentyfold for an hour (*False Missile Alert* F-7). And a regional disaster floods every center in the region at once: Houston's emergency center received some 56,000 calls for service in fifteen hours during Hurricane Harvey, against about eight thousand in an ordinary period of that length, and calls for rescue waited hours and then a day (*Hurricane Harvey* §3.5).
Consequence threshold callers with real emergencies waiting beyond the center's answering standard, or abandoning the call, because positions are occupied by other calls.
Required function the center must be able to overflow calls to alternate centers before its own queue reaches the point at which real emergencies wait too long, and must be able to see the pattern of a generated flood — its behavior — short or silent calls, hang-ups, immediate redial, and a rate of such calls above the center's baseline — rather than the identity of the callers, which a flood from many real phones or from randomized identifiers does not share (*Call Flood* F-2) — quickly enough to act on it. A center cannot refuse a 911 call. The design basis therefore does not assume calls can be blocked; it requires that capacity be added by overflow and that the center be able to triage the queue it has. Overflow must reach centers outside the region a disaster strikes, for a natural event as for a coordinated flood (Section 5.4). In a deliberate mass-casualty incident the center must also be able to vet reports of further incidents before units are committed: recording each with its location, caller and time, relating it to what is known of the true scene, and passing it to command for decision. In Las Vegas such reports were echoes of the dispersing injured, which command vetted before committing resources, and under military-action conditions an adversary may add reports designed to survive that vetting (*Route 91 Harvest Shooting* F-2, F-7; *Response Forces and Stations* §5.3). The center's plan for a planned gathering above the area's threshold includes a staffed dispatch position for the event; in Las

Vegas the special-events channel had none (*Response Forces and Stations* §4.7). Sorting the queue is an action placed on its consequence: a real caller sorted down it waits, and automated sorting that can reach the threshold falls under the conditions of Section 6.4. The time to remove a flood's sources — by the law-enforcement, hosting and platform actors who can take down the means of inducement — is part of the flood's design duration, and where it cannot be bounded the center plans its endurance under flood for the longest plausible duration (*Call Flood* F-3). The answer to a generated flood depends on an obligation the class does not own: that a device able to call 911 does so only on a human act at the point of origin (*Access and Edge*; *Call Flood* F-1). And in a regional disaster the area considers a line separate from 911 for disaster information, which takes from the center the calls that are not emergencies, as the Maui police after-action report recommended (*Lahaina Wildfire* R-7).

Design-basis duration the duration of the surge, which for a disaster is hours to days and for a generated flood lasts until its source is stopped.

4.4 Loss of Computer-Aided Dispatch

Postulated event loss of computer-aided dispatch through hardware failure, software fault, loss of the hosted service or network that carries it, or a cyber attack.

Consequence threshold a center that cannot record incidents, track units or select which units to send, and so dispatches late, twice or to the wrong place.

Required function every center must be able to dispatch by hand, with paper incident records, printed response plans and maps, and a unit status board, and must practise doing so. Computer-aided dispatch must not be a single point of failure for answering: the call-handling equipment and the radio consoles must keep working when dispatch software does not. Baltimore in 2018 and Dallas in 2023 kept dispatching because their staff could work without the system; the design basis requires that this be shown, not assumed. The separation of call handling and radio from dispatch software is required by design, not credited because it happened to hold; Baltimore and Dallas are the practice references (*Baltimore Dispatch* F-2; *Dallas Ransomware* F-2). The manual mode covers each function lost with the software, not the software alone: alerting stations by radio or telephone, updating units by voice in place of their screens, locating addresses without the city's map, and holding the caution and prior history of addresses in a form that survives the loss of the records system; in Dallas the units' mobile data and the stations' alerting went down with dispatch (*Dallas Ransomware* F-3). During manual operation the incident record is kept on paper in a form entered on restoration, so that the record has no gap (*Baltimore Dispatch* F-4). And on the author's decision of 7 October 2026, the showing this section requires is recorded evidence: the times of manual call handling and dispatch, and the load on the radio, measured against ordinary operation in each exercise under CDB-7 and in any real event, and recorded rather than made a class parameter. Neither Baltimore nor Dallas published such a measure (*Baltimore Dispatch* F-4; *Dallas Ransomware* F-4).

Design-basis duration the time to restore the system, which after a cyber attack has ranged from hours to days or longer. Where an attack reaches the network dispatch depends on, the area states a planning duration for manual operation, with the staffing, supplies and rest it needs, and plans to restore dispatch by rebuild from backups the attacker could not reach; Dallas returned to full automation eight days after its attack. On the author's decision of 7 October 2026 the rule is stated and the value left open (Section 10.1; *Dallas Ransomware* F-5). The planning duration of manual operation is stated in staff as well as in time, since dispatch by hand takes more of each

telecommunicator's time, and it is counted against the sector's record of staff on hand (*Emergency Services Sector Framework* §3.5; DBA-ESS-UC-2).

4.5 Loss of the Center Building

Postulated event fire, flood, structural failure, contamination or a threat that forces evacuation of the center.
Consequence threshold loss of every position in the center at once.
Required function every center must have a backup center, or a tested arrangement with another center, that does not share its building, its flood or fire exposure, its power supply or its communications entry. The backup must receive calls with location for the whole area and see the area's units, within the transfer time. A center without a backup of its own must hold a written and exercised arrangement with a neighbor that meets the same test.
Design-basis duration the time to return to the primary center or to establish operation elsewhere, which may be days to months.

4.6 Loss of Public-Safety Radio

Postulated event failure of a radio site, of the controller that links the sites of a trunked system, of the links between the consoles and the sites, or of the timing on which simulcast transmission depends.
Consequence threshold units in an area out of contact with the center and with each other.
Required function radio sites must fall back to operating on their own when they lose the controller or their links, so that units near a site can still talk through it. The center must hold fallback channels, including mutual-aid channels, that do not depend on the trunked system, and consoles that can reach a site directly. Simulcast systems must state how long they keep transmitting correctly if their timing signal is lost, and their timing receivers take their design basis from DBA-COMM-PN-FC4 (*Positioning, Navigation and Timing*) §7. On the author's decision of 12 October 2026, from DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*), the users the radio system serves under denial include each receiving hospital whose named path to the area's command is a terminal on the system (*Care Delivery* §4.3). The system's design basis counts those terminals among the users it must carry when its sites operate on their own and when traffic moves to mutual-aid channels, and states for each such hospital the site or channel by which it remains reachable in that condition, so that the count of patients by hospital that command at the scene must hold (*Response Forces and Stations* §4.7) still reaches command. A hospital reachable only through the trunked system's controller, or through backhaul the same event can cut, is not credited as reachable under denial. The terminal, its power from the hospital's essential electrical system and the people trained to work it remain the hospital's design basis (*Care Delivery* §4.3).
Design-basis duration the time to restore the failed element, and for timing, the holdover period.

4.7 Cyber Attack on Call Handling, Dispatch and the 911 Network

Postulated event compromise of the call-handling equipment, computer-aided dispatch, the radio system's management, or the Next-Generation 911 network and its core services, including through the vendors and remote-access channels that maintain them.
Consequence threshold a single compromise that takes answering, dispatch or radio out of service across one or more centers, or that corrupts what telecommunicators see.

Required function the call-handling, dispatch, radio and network systems must be separated so that the compromise of one does not take the others with it, and answering must survive the loss of dispatch. A change that can reach many centers on a shared network or a shared hosted system must not be applied to all of them in one operation without a governed act, meaning an authorized and recorded change made for each group of centers the change could take out together (*Local Sovereignty* §3.2). Backups of dispatch data must be kept where an attacker who reaches the live system cannot reach them. Separation includes identity and administration: these systems do not trust the accounts of the enterprise domain, are not reachable by its remote-management tools, are administered through paths a compromise of the enterprise does not reach, and are monitored for an intruder who uses legitimate tools; in Dallas the attackers reached dispatch with an ordinary service account and third-party remote-management tools and stayed about four weeks (*Dallas Ransomware* F-1). Any change to the protection of one of these systems — a firewall rule, a remote-access path, an exception made to troubleshoot — is a governed act at a single center as well as across many: authorized and recorded, checked for exposure once made, and set to expire if temporary; Baltimore's opening was a firewall change made while fixing something else (*Baltimore Dispatch* F-1). The requirement holds whether or not an attacker selects dispatch, since an exposed system is found by attackers looking for anything (*Baltimore Dispatch* F-5). And the data of dispatch — unit assignments, station alerting, response plans and address histories — is protected against disclosure as well as alteration (*Dallas Ransomware* F-6). A backup out of the attacker's reach is not thereby a backup the attacker did not corrupt, since an intruder who stays for weeks may alter the data before it is backed up (Dallas Ransomware F-1). A restored system is checked, before it is used, against the approved copy of the response plans, protocols and gatekeeper settings that Section 6.4 requires and against a roster of units held outside it, and the paper record of manual operation is reconciled with it (DBA-ESS-UC-2).

Design-basis duration indefinite; the condition lasts until the compromise is found and removed.

Table 4-1. Class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Measure
Power loss	Grid outage outlasting batteries and fuel; staff not relieved	Chain A, Chain F	Operate-through duration
Upstream delivery failure	Carrier, router, routing center or core services fail	Chain A	Alternate-answering transfer time
Volume beyond capacity	Disaster surge or generated flood of calls	Chain A	Transfer time (overflow)
Dispatch software loss	Fault, hosted-service loss or attack	Chain D	Operate-through duration
Center building loss	Fire, flood, evacuation	Chain A, Chain D	Transfer time; alternate-routing map
Radio loss	Site, controller, link or timing failure	Chain F	Operate-through duration
Cyber attack	Compromise of call handling, dispatch, radio or 911 network	Chain A, D or F	Centers reachable by one act

5. Military-Action Extension: DBA-MA for Answering and Dispatch

5.1 MA Extension Applicability Determination

Every 911 center, backup center and public-safety radio system of consequence performs the facility-level applicability determination the series makes mandatory (framework §7), under the four-filter applicability gate of DBA-MA-AF (*MA Applicability Framework*) §2 and §3. Any one filter satisfied brings the facility into scope. For this class the filters are met most often by adversary targeting logic and by geolocation relative to defense-critical loads: a center that answers for a military installation, a port, a grid control center or a dense population is a target because of what its loss does to every other response in its area. The framework records a strong presumption toward applicability for 911 centers and public-safety communications (framework §7). The determination is recorded, and any scope reduction carries a recorded basis (Section 5.8).

5.2 The Four Characteristics Applied to the Class

Applied to answering and dispatch, the four military-action characteristics take specific form. Intentionality: the adversary chooses the 911 centers whose loss leaves the largest population without help, and times the act to the moment help is most needed. Persistence: the adversary pre-positions in dispatch systems, call-handling equipment or the vendor channels that maintain them, and waits. Precision: the adversary takes out the center, its backup and the path between them, and leaves the rest of the network working, so that calls appear to go through while no one answers them. Denial: the adversary floods the center with generated calls, cuts its power, and attacks the roads and fuel on which its endurance depends, so that answering, transfer and recovery are denied together. The class validation standard is the state actor who knows that degrading the response multiplies the consequence of a primary attack on another sector (framework §6.2, §7).

5.3 Dispatch Denied During a Primary Attack as the Governing MA Scenario

The governing scenario is a compound, second-wave one. A primary attack strikes another sector — the grid, a chemical plant, a water system — and produces the surge of calls that such an attack produces. At the same time the area’s 911 centers lose commercial power, their call delivery is degraded, their dispatch software is disabled, and a flood of generated calls occupies their positions. It joins the five reference events: the derecho’s power loss, the 2014 outage’s silent failure of call delivery, the 2016 flood, and Baltimore’s and Dallas’s loss of dispatch, combined by an adversary who intends all of them at once.

The scenario sets the class’s adequacy test. An area passes if, for the design duration of the compound event, calls for the area keep reaching a staffed position with enough location to send help, within the transfer time set for the area; responses keep being directed, by hand if need be; and units keep being reachable by radio. An area that cannot pass carries a declared residual (Section 5.7).

5.4 The Coordinated Call Flood

The second scenario uses the volume case deliberately. The adversary controls or induces enough devices to call 911 at once, across several areas, timed to a primary attack. Because a 911 center cannot refuse a call, the flood consumes capacity that real callers need, and overflow to neighbors moves the problem rather than ending it if the neighbors are flooded too. The class answer is the requirement of Section 4.3, stated against a coordinated flood: overflow must reach centers outside the flooded region, the center must be able to recognize and sort a generated flood quickly, and the alternate-routing map must name alternates far enough away that one flood does not reach them all. A flood need not occupy the centers' positions to deny answering. If it can exhaust a finite resource upstream — a counter, a table, a license limit — it stops calls before they reach any center, silently, and the monitoring rates the result as minor. That is the route the 2014 outage showed by accident; the requirement of Section 4.2 that every finite resource report its margin is its answer (*Multistate 911 Outage* F-8). An induced flood needs no devices under the adversary's control: each call comes from a phone its own user was induced to make, and the measure is distribution — how many people the means of inducement reaches, and how densely in each answering area. The flooded region is drawn by the audience, not the geography, and the alternate-routing map names overflow alternates accordingly (*Call Flood* F-4).

5.5 The Dispatch System as a Target

The third scenario uses Chain D. The adversary gains access to computer-aided dispatch, or to the hosted service that runs it, and disables or corrupts it at a chosen moment. Corruption is the worse case: a dispatch system that is down is obvious, while one that shows wrong unit locations or recommends the wrong units is believed. The class answer is the requirement of Section 4.4 — every center can dispatch by hand, and practises it — together with two further requirements. Telecommunicators must be able to tell when the system's picture of the units disagrees with what the units report by radio, and must have the authority to stop using it. And the indications on which any recommendation rests must satisfy *Indication Integrity*, so that a recommendation built on corrupted unit locations is not acted on as if it were sound (Section 6.4). The 911 authority names, before an incident, who may take a dispatch system offline on suspicion of compromise, on what indication, and how the center is told, so that the decision can be made in minutes; in Baltimore the city's and the vendor's staff took the server down. Because a corrupted system gives less reason to suspect it than a failed one, the indications include disagreement between the system's picture and what units report (*Baltimore Dispatch* F-3).

On the author's decision of 11 October 2026, two further requirements apply, from DBA-ESS-UC-2. Detection by disagreement rests on the radio a compound event degrades, so the comparison does not rest on voice radio alone: each unit reports its position and status from the unit, authenticated there, over a path that does not pass through the dispatch system, and that report is compared with what the dispatch system shows (*Emergency Services Sector Framework* §4.2). And under the presumption of Section 5.1, an unexplained disagreement between the

dispatch system and the units during a primary attack is in itself a sufficient indication for the named official to take dispatch offline, stated in advance, because a corruption designed to be believed will not supply proof.

5.6 The Adequacy Result: Surviving Answering Condition

The test of whether an area's answering and dispatch protection is adequate is whether it can reach what this document calls the Surviving Answering Condition. When the grid is lost, call delivery is degraded and outside support cannot arrive, calls for the area still reach a staffed position with enough location to send help, within the area's transfer time; responses are still directed, by hand if need be; units are still reachable by radio; and this holds for the design duration of the compound event.

The protective functions on which this condition rests are the automatic transfer of a center's and a radio site's load to battery and the automatic start of their generators when grid power fails, and the fall-back of a radio site to operating on its own when it loses its controller. They are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2). Each decision is a measurement at the facility — a loss of voltage, a loss of the controller link — against a setpoint; the function is colocated with the facility it protects; no system an attacker could reach can stop, delay or trigger its action; and no such system can alter its setpoints, logic or bypass state except through a governed act. *Local Sovereignty* governs and is not restated here. A radio site whose fall-back can be disabled from a management system an attacker can reach fails that test. Local sovereignty describes what protects the facility when its management systems and the authorization architecture around them have failed. It does not decide whether any action needs human authorization (*Local Sovereignty* §4).

Each center, backup center and radio site must also hold its function for a set period with no outside power, communications or support. That requirement is written here in this document's own words. Other instruments take a similar requirement from the DBA-MA parent framework's "local sovereignty principle", but whether that principle is the same thing FD-LS defines is still an open question, so this document does not borrow it (Section 10.4).

5.7 Adequacy and the Declared Residual

In the non-adversarial domain, the Surviving Answering Condition is achievable with the Section 4 requirements: endurance matched to the design duration, a backup that shares nothing with the primary, recognition of silent failure, overflow to alternates, and dispatch by hand. Against a deliberate attacker, adequacy has limits.

Two residuals are declared, in the output form of DBA-MA-DiD (*Defense in Depth*) §3.5. First, a failure of call delivery across every carrier serving an area, upstream of every 911 center, cannot be repaired by the centers. Beyond the point at which no alternate route survives, the residual is that people in the area cannot reach 911 by telephone, and the response lies with the coordination class: public instruction, by channels that do not depend on the telephone network, to go to staffed fire stations, police posts or other points where help can be summoned in person

(DBA-ESS-EMCo, framework §3.4). Second, a coordinated flood of generated calls large enough to occupy every alternate within reach cannot be prevented by the centers, which may not refuse a call. The residual is delay for real callers for the duration of the flood, and the response is the sorting of Section 5.4 and the pursuit of the flood's source by others. These residuals bound the class adequacy claim. A third residual is declared on the author's decision of 11 October 2026: the interval between a corruption of dispatch and its recognition, during which responses are misdirected on a system the center has no reason to doubt. It is bounded by the interval of the checks of Sections 5.5 and 6.4 and cannot be removed (DBA-ESS-UC-2). The first residual is handed to the coordination class, and under the framework's principle that a residual is held where it is handed it is held only if that class can show it holds it; in the compound event it relies on stations the same event is emptying, and the framework declares it at the sector level (*Emergency Services Sector Framework* §4.2; DBA-ESS-UC-1).

5.8 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a shorter design duration accepted, a backup center that shares a building, power supply or communications entry with the primary, a longer transfer time for a rural area, a center credited with dispatch by hand that has not practised it — must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Section 10.1 records it for the framework to adopt.

6. The Dual Design Basis, Stated in Full

6.1 The Two Design Bases

A 911 center carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 6.5.

6.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions a center relies on to keep answering and dispatching independently of the systems an attacker could reach. These are the transfer to battery and the start of the generator at the center and its radio sites, the fall-back of radio sites to operating on their own, the failover of call-handling equipment to its backup, and the integrity of the indications telecommunicators see. Power transfer is the clearest locally sovereign function in the class when it is built so: the decision is a loss of voltage measured at the facility, and the switch acts there.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Power transfer, radio fall-back and call-

handling failover that behave predictably within a stated range of inputs, and that have been formally tested as installed, are exempt from the Command Broker requirement for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not thereby placed below the Bright Line: they decide whether a center keeps answering, and they are outside the obligation because their behavior is bounded by proof (*The Bright Line* §4.5). The same holds for a medical dispatch protocol implemented as fixed question-and-answer logic: if it is deterministic in that sense, it is outside the scope of the obligation, and that says nothing about the consequence of the determinations it produces. A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence.

6.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are arriving in answering and dispatch in five forms: systems that transcribe, translate and summarize calls; systems that score the priority of a call or suggest its medical determinant; dispatch systems that recommend which units to send; systems that recommend moving units between stations to keep coverage, often called move-up or system status management; and systems that dispatch certain call types automatically. They are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. This document names the actions in answering and dispatch that the architecture applies to. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Four cases illustrate the application, and none of them is a placement by component. A dispatch recommendation for a cardiac arrest or a structure fire, whose error would delay the response at or above the severity threshold, is above the line; a recommendation of which of several equivalent units should take a low-priority report, whose consequence cannot reach that point, is placed on that consequence and may fall below it. A priority score that reorders the queue of waiting calls takes one action with each reordering, and a reordering that could leave a life-threatening call waiting behind others is above the line. A move-up that strips an area of its last available unit is above the line, whatever the size of the move. And an automatic dispatch takes the action itself, with no person in the path, so each one is placed on its consequence and, where above the line, can be taken only under an authorization given in advance under Section 6.4.

Whether the obligation reaches an action depends on how the system is built. Where a telecommunicator reads a system's output and then decides and acts herself, the action is hers, and the output carries the obligation of *Indication Integrity* rather than the authorization obligation (*Mode of Discharge* §7). Where the system's output would itself make the change — a recommendation that dispatches the units the moment it is accepted with one keystroke — the obligation reaches it, because the output makes the change through its acceptance (*The Bright*

Line §4.3; *Mode of Discharge* §7). The difference lies in the design of the dispatch screen, and a supplier must declare, for each output, which of the two it is.

Consistent with CDB-6 of the Military Action Communications Design Basis, the loss or degradation of the center's own communications does not suspend the Command Broker obligation. CDB-6 states the obligation for actions recommended by generative AI; this instrument applies it to every component within the scope of *Layer Decomposition and Determinism* §4.3, which is the scope the Foundational Definitions set.

6.4 Authorization in Seconds: The Pre-Authorized Envelope

Dispatch happens in seconds, and a telecommunicator handling a cardiac arrest cannot also deliberate over each recommendation a system makes. On the author's decision of 6 October 2026, authorization for a bounded class of dispatch actions may be given in advance: by the medical director for medical dispatch, and by the responding agency for its response plans, with every action outside that class referred to a person in real time. This is advance discharge under *Mode of Discharge*, and the pre-authorized envelope is the authorization type that *Pre-Approved Action Envelope* (CB-PAE) defines in the Command Broker family. Neither document is restated here. This section states how they apply to answering and dispatch, which is the first sector instantiation of advance discharge in the emergency services series. The action remains above the Bright Line; what moves is the time at which it is authorized, not whether it is (*Mode of Discharge* §2.2, §4.4).

Advance discharge is available only where its conditions are met. A verified gatekeeper that the recommending system cannot modify or bypass must admit each action, and its acceptance set must be proven to bound not only whether an action is admissible but which action is selected, over state, trajectory and rate (*Mode of Discharge* §4.1, §4.2). Applied to dispatch, this divides the cases. A unit recommendation inside an approved response plan, checked against measured unit status, unit location and capability, can in principle be admitted by such a gatekeeper. The composition clause of *Mode of Discharge* §4.2 has a direct dispatch meaning: a sequence of admissible assignments must not be able to strip an area of coverage, so the gatekeeper must hold a coverage floor as well as the response plan. A score that ranks the queue of waiting calls, or a machine-learned suggestion of a medical determinant, is a selection that no such proof now bounds. Where the proof cannot be produced, advance discharge is not available, and the action takes in-the-moment authorization by the telecommunicator or the floor supervisor (*Mode of Discharge* §4.2). That is the honest finding of this section: the pre-authorized envelope covers the part of dispatch that is already rule-bound, and the newest systems fall outside it until their selections can be proven.

Automatic failover of call routing is an action of the same kind, placed on its consequence. It moves the 911 traffic of every area an element serves, and a failover that misroutes a state's calls, or moves them to a center that cannot take them, is as large a harm as the outage it answers. Placed on that consequence, a failover rule for an element whose traffic reaches the

severity threshold is above the line. It fits advance discharge closely: a deterministic rule, a measured trigger such as the completion rate, an acceptance set that bounds where traffic may be moved and when, and a verified gatekeeper. The approving official is named in the alternate-routing map. Where the element is contracted, the official is the 911 authority's when the element falls on this class's side under the contract test of Section 8.3, and the provider's under the communications series where it does not (*Multistate 911 Outage* F-4).

Mode of Discharge §7 requires each sector to instantiate three things, and this class does so as follows.

Denied-condition immutability. When computer-aided dispatch, its unit-status or location inputs, or the network that carries them are degraded, or a compromise is suspected, the pre-authorized envelope becomes immutable: no change to a response plan or to the gatekeeper is made until a deliberate, authenticated act restores normal operation. Where the inputs no longer satisfy *Indication Integrity*, advance discharge is not valid for any action that rests on them, and dispatch reverts to the telecommunicator, by hand if need be. That reversion is the operate-through mode Baltimore and Dallas exercised, and the class requires it to be ready at all times (Section 4.4). On the author's decision of 11 October 2026 the envelope is also frozen when a deliberate attack is in progress in the area it serves, as for a suspected compromise (*Emergency Services Sector Framework* §5), and while the attacker is unlocated no automated system dispatches to a reported deliberate attack. The holder of the decision to send units to such a report is named in the area's plan for military-action conditions, with *Response Forces and Stations* §5.3 (DBA-ESS-UC-3).

Reach of the obligation preserved. An above-line dispatch action that a recommending system initiated or materially influenced is within the obligation's reach, including an action the system makes through a telecommunicator's one-keystroke acceptance (Section 6.3). Advance discharge does not narrow that reach.

Qualification for the advance mode. The medical director who approves medical dispatch protocols and their response assignments, and the agency official who approves response plans, are the Command Broker discharging in advance when those approvals form a pre-authorized envelope, and must be qualified for that mode as well as for their clinical or operational roles (*Mode of Discharge* §7). Accountability rests with them and with the 911 authority that operates the center (*Mode of Discharge* §8).

The envelope needs no new governance apparatus, because 911 centers already run the machinery *Pre-Approved Action Envelope* §3 looks for. Medical dispatch protocols are approved by a medical director and changed under control; response plans are approved by the responding agencies and maintained as controlled tables in the dispatch system. These are the emergency services counterparts of the change-control regimes that document names for other sectors (*Pre-Approved Action Envelope* §3.5). A change to a response plan, a protocol or the gatekeeper is a change to the envelope, and is excluded from advance discharge (*Mode of Discharge* §4.3). And the hazard *Pre-Approved Action Envelope* §4.2 names — an optimizer that moves to the edge of its bound and stays there — has a dispatch form in move-up systems that hold coverage at its floor to save driving time. Sustained operation at the coverage floor is a re-validation trigger (*Pre-Approved Action Envelope* §4.3).

An envelope that cannot be shown to be the one approved cannot support advance discharge. A change made by an adversary is not made through the act *Mode of Discharge* §4.3 assumes, and no compromise is suspected to freeze it. The response plans, protocols and gatekeeper settings in use are therefore compared with an approved copy held outside the dispatch system, at a set interval and before any return to automation, and a mismatch is an indication of compromise under Section 5.5. Returning to automation after dispatch has been taken offline is above the line on its consequence and is authorized by the officials who approved the envelope as well as by the 911 authority (DBA-ESS-UC-2).

6.5 The Interaction Point: Answering Under Denial

The two design bases meet when a center must keep answering and dispatching while under attack. The control system must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization, given in the moment or in advance under Section 6.4. That includes any dispatch, reordering of the queue, move-up or automatic dispatch whose consequence reaches the threshold. And the center's power, its radio sites and its ability to answer and to dispatch by hand must keep working even when no outside system or authority can be reached at all. The physical design basis guarantees that the center keeps answering; the AI-governance design basis guarantees that no recommending system can misdirect its response without a qualified human authorization the architecture can enforce.

6.6 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, and the Command Broker family under its register. Where the two are cited together, the physical requirement governs the center's protective architecture and the AI-governance requirement governs the authorization of any recommending system that interacts with it.

7. The Communications Design Basis Applied to Answering and Dispatch

7.1 Why It Applies

A 911 center receives other people's communications and depends on communications of its own. It reaches its backup center, its neighbors, its units, hospitals and emergency managers over links that may fail in the same event that floods it with calls. The Military Action Communications Design Basis applies to those links as it applies to any facility (*Military Action Communications Design Basis* §3). This section states how each of the seven requirements applies to answering and dispatch. It cites them; their text is in that document.

7.2 The Finding That Shapes the Rest: A Silent Center Looks Like a Quiet Night

When calls stop arriving because the network that delivers them has failed, a 911 center sees nothing wrong. The phones are quiet, and a quiet phone is what a good night looks like. The 2014 outage lasted six hours for that reason. A center that depends on the failed network to tell it the network has failed will not be told. The requirements below follow from this finding. On the author's decision of 11 October 2026 the framework states the answer for the sector: every answering point and radio site sends an authenticated heartbeat to the operations center at a set interval, on a channel independent of what it reports on, so that its silence is itself the alarm; the interval bounds the recognition time counted in the transfer time of Section 3.2 (*Emergency Services Sector Framework* §4.2).

7.3 The Seven Requirements

CDB-1, degradation as baseline. For this class the degraded condition is the ordinary one (framework §1.5). The design basis assumes from the outset that the center's links to its backup, its neighbors, its units and the networks that deliver its calls are degraded or lost in any event large enough to matter, and the transfer, overflow and manual dispatch plans are written and tested against that condition.

CDB-2, local protective and safety autonomy. This requirement is still worded for the electric sector, where it names a grid's supervisory control and energy-management systems. Read for answering and dispatch, it requires that loss of the dispatch system, the hosted service or the 911 network not cause loss of the center's ability to answer the calls that do arrive, to talk to its units by radio, or to dispatch by hand. Section 5.6 states the property, and the design basis must demonstrate it rather than assume it.

CDB-3, field-executable emergency procedures. Telecommunicators must be able to dispatch by hand, move to the backup center, invoke alternate routing and fall back to mutual-aid radio channels from procedures, response plans and maps held in a form that does not depend on the network.

CDB-4, verified independent channels. Each center must be able to reach its backup center, the centers named in its alternate-routing map, and the providers that route its calls over a channel that shares nothing with its primary links. It must also be able to learn, by a means independent of the delivery network, that calls have stopped arriving (Section 7.2). Notification of centers by the operator of a failing element is triggered by the consequence-rated detection of Section 4.2, so that it does not wait on a person's recognition. The detection on which notification rests is powered and connected independently of the element it watches (Section 4.2).

CDB-5, coordination without shared connectivity. This requirement is also still worded for the electric sector, where it names the gas–electric boundary. Read for answering and dispatch, it requires that neighboring centers, the 911 authority, the carriers and service providers that deliver calls, and the emergency operations center be able to coordinate transfer, overflow and

rerouting when their usual channels are down. Procedure and contract alone have twice failed to meet this requirement for notification: in 2014, and again in the outage of September 2020 after six years of compliance plans (*Multistate 911 Outage* F-7).

CDB-6, Command Broker continuity. Sections 6.3 and 6.4 apply it.

CDB-7, verification cadence. Transfer to the backup center and to the alternates of the alternate-routing map is tested at set intervals with the primary links assumed lost, and the test includes a period of dispatch by hand. Each test measures the times of manual call handling and dispatch against ordinary operation and records them (Section 4.4).

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On: The Mitigation Layer

Every sector depends on this class, because every sector's failure ends in calls to 911. The framework states the outward consequence of the sector as a consequence multiplier: degraded response amplifies every other sector's failure (framework §6.2). For this class the multiplier is literal. A chemical release, a water contamination, a grid outage or a building collapse is answered, or not, through a 911 center, and the center's loss turns a contained incident into an uncontained one. The class therefore holds its tightened criteria for its own sake rather than at another sector's request.

8.2 What Answering and Dispatch Depends On

The class depends first on the communications sector for the delivery of calls and location, and that dependency is where the critical-services overlay of the communications series takes effect: an access facility that is the only path for a 911 center inherits the tightened criterion of *Access and Edge* §8.1. It depends on electric power and on the fuel and roads that keep its generators running (framework §6.1). It depends on timing, for the time stamps on its records and for simulcast radio transmission, and its timing receivers take their design basis from *Positioning, Navigation and Timing* §7. It depends on the facilities that host its dispatch system and network core when those are hosted elsewhere; their building-level protection is the data-center series' and is cited there, not restated. And it depends on its telecommunicators, whose numbers, endurance and training set the ceiling on everything else in this document. It depends, for the end of a generated flood, on the law-enforcement, hosting and platform actors who can remove its sources, and on the obligation that a device call 911 only on a human act at the point of origin (Section 4.3).

8.3 Boundaries

With Communications. On the author's decisions of 6 October 2026, the boundary lies where the call and the caller's location enter the 911 system. In an enhanced 911 system, the 911 trunks, selective routing and location delivery are operated by a carrier or a 911 service provider and belong to the communications sector, and the 911 system begins at the center. In a Next-

Generation 911 system, the Emergency Services IP Network and its core services are owned by the 911 authority and belong to this class, and the 911 system begins where the originating network hands the call to that network. The boundary is the same rule in both cases; it lies further upstream in the second because the 911 authority owns more. A contracted function is the 911 authority's when the contract binds the provider to this class's design obligations for that function — detection, an independent alternate, invocation within the transfer time and notification — and the authority can enforce them; on the author's decision of 7 October 2026, this contract test replaces the rule that a core service run for the authority by a contractor remains the authority's, which did not say how to read the contract. A function that fails the test is on the communications side. On the public record, the Washington routing element of the 2014 outage failed it (*Multistate 911 Outage* F-2). Whichever side an element falls on, the design obligations of this class for the call path follow the call: they are carried by the communications series' classes on that side and by contract where the authority buys the service. On the author's confirmation of 7 October 2026, the core services of a Next-Generation 911 system sit with the Emergency Services IP Network on this side: they run on that network and are owned by the same authority. Where the contract test places a function on the authority's side, the contract binds the provider to supply the evidence of backup power, diversity and monitoring that Section 3.4 asks the map to hold.

With FirstNet. FirstNet is a carrier network with priority for public safety, and belongs to DBA-COMM-AC-FC2. A center or unit that depends on it as a sole path holds that dependency under the elevation rule of the communications series.

With the critical-services overlay. The communications series carries 911, the Emergency Alert System and the priority services as an overlay (Communications Sector Framework §3.7). This class takes the facilities on its side of the boundary and leaves the overlay the elevation of the communications facilities on the other side. Section 10.3 records what that means for the overlay.

With Response Forces. The units in the field, their stations and the radios they carry belong to DBA-ESS-RF-FC2. The radio system they talk through, from the sites to the consoles, belongs here.

With Emergency Management. Emergency operations centers, incident command and the origination of public alerts through the federal alerting gateway belong to DBA-ESS-EMCo. The center feeds them and takes direction from them during a large incident.

With Healthcare. Medical dispatch protocols are approved by a medical director, whose clinical authority belongs to the healthcare sector at the patient-transfer boundary (framework §6.3). The design basis of the dispatch action, and of the center that takes it, belongs here, even where the person who approves it in advance holds a clinical qualification (Section 10.8). A receiving hospital whose path to command under denial is a terminal on the area's public-safety radio is a user of the system this class governs, and Section 4.6 counts it among the users that system serves; the terminal itself is the hospital's.

8.4 Regulatory and Institutional Context

The Federal Communications Commission sets requirements for the reliability of 911 service providers, adopted after the derecho, covering circuit diversity, central-office backup power and network monitoring — certified each year until June 2026, when the Commission decided to replace the annual certification with a one-time certification updated on material change, after an eighteen-month transition, and extended the rule to Next-Generation 911 providers — and requires providers to notify 911 centers of outages that affect them. It sets the enhanced 911 location requirements and the framework for Next-Generation 911. States and their 911 boards fund and oversee the centers and, increasingly, own their Next-Generation 911 networks. The National Emergency Number Association publishes the standards for call answering and Next-Generation 911 that centers follow, and the National Fire Protection Association publishes the standard for emergency communications centers. The Federal Emergency Management Agency provides the doctrine for interoperable communications and runs the federal alerting gateway. P25 radio follows the TIA-102 suite of standards. None of these sets an engineering design basis for how long a center must keep answering through a compound failure, or how quickly its calls must reach another center when it cannot. That is the gap this class fills (framework §2.3).

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

Table 9-1. Inheritance map for DBA-ESS-AD-FC1.

Element	Source	Inheritance mode
Function-primary taxonomy and the class's place in it	framework §3.1, §3.2, §8.2	Inherited unchanged
Operate-through requirement as a defining characteristic	framework §1.5, §3.5	Inherited and developed (Sections 2.1, 3.2)
Consequence-severity inclusion criterion and candidate events	framework §4.1	Inherited and developed (Section 4)
Bright Line and Command Broker architecture	framework §5; FD-BL; FD-BR	Applied; not restated (Section 6)
Determinism as scope	FD-LD §4.3	Applied (Section 6.2)
Advance discharge and its sector instantiation	FD-BL-D1 §4, §7, §8; CB-PAE §3, §4	Applied (Section 6.4)
Indication integrity	FD-BL-D2	Cited (Sections 5.5, 6.3, 6.4)
Envelope reservation	FD-EV	Applied (Section 1.5)
Local sovereignty of power transfer and radio fall-back	FD-LS §3.2	Cited directly (Section 5.6)
Communications design basis CDB-1 to CDB-7	DBA-MA-CDB §2, §3	Cited and applied (Section 7)
MA applicability gate	DBA-MA-AF §2, §3	Applied (Section 5.1)
Declared residual	DBA-MA-DiD §3.5	Cited directly (Section 5.7)

Element	Source	Inheritance mode
Critical-services elevation at the access boundary	Access and Edge §8.1	Cited (Section 8.2)
Timing receiver design basis	Positioning, Navigation and Timing §7	Cited (Sections 4.6, 8.2)
Boundary with Communications	author's decisions of 6 October 2026	Stated here (Sections 1.2, 8.3)
Operate-through duration, alternate-answering transfer time, alternate-routing map, Surviving Answering Condition	—	Defined here

9.2 Anticipated Derivatives

The class anticipates two kinds of derivative. A use case for the governing scenario of Section 5.3, applied to a named county with a defined compound event, would test the Surviving Answering Condition and the alternate-routing map with numbers. Reference-event instruments for the five conferred events would record what is established about each; the 2014 outage, for which the Federal Communications Commission's report gives a full account of the failure and of the six hours before transfer, is the natural first.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

Operate-through duration, alternate-answering transfer time and the alternate-routing map are stated as parameters, on the author's decision of 6 October 2026. The transfer time required for an area, and how it should differ between a dense city and a rural county, is still to be set, and the framework should adopt the conditionality principle of Section 5.8 for the sector. The recognition time each area sets as part of its transfer time is also to be set (Section 3.2), and the framework should adopt the recording of concentration (Section 3.4) at the sector level. On the author's decision of 7 October 2026 two further values are left open with their rules stated: the planning duration of manual operation after a cyber attack on the network dispatch depends on (Section 4.4), and the cadence on which the area renews its evidence of the upstream elements (Section 3.4).

10.2 Design-Basis Durations

The design duration of the compound event is the duration that governs the class, and no figure is given here. It should be set together with the energy series' assumptions about restoration after a deliberate attack on the grid and with the communications series' design outage for access, so that the three series assume the same event.

10.3 The Boundary with Communications and the Critical-Services Overlay

This class was written first in order to learn what the communications series' critical-services overlay must still govern, and the answer is now visible. The overlay names three services. For 911, this class has taken every facility on its side of the boundary, and what is left to the

communications series is the elevation of the networks that deliver calls and location — which *Access and Edge* §8.1 already states. As Next-Generation 911 spreads, the boundary moves upstream and the communications series' share of 911 shrinks further. For alerting, origination belongs to this series' coordination class and broadcast to *Access and Edge*. The priority services are wholly the communications series', and no facility belongs to them. No facility is left for the overlay to own. This supports keeping it as an elevation overlay rather than making it a facility class. On the author's decision of 6 October 2026 the overlay stays an elevation overlay with no instrument of its own, Communications Sector Framework §3.7 states it, and Project 9 OI-24 is closed.

10.4 Local Sovereignty in the DBA-MA Sense

The protective functions at Section 5.6 cite *Local Sovereignty* §3.2 directly. The endurance requirement beside them is the property the DBA-MA parent framework's local sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is written here in this document's own words until then. The item is registered under Project 9 OI-16.

10.5 Reference Events

The derecho of June 2012, the multistate outage of April 2014, the call flood of October 2016, and the ransomware attacks on Baltimore in March 2018 and Dallas in May 2023 are conferred by the framework at its §9.1 as DBA-ESS-RE-1 through DBA-ESS-RE-5, and all five instruments were written on 7 October 2026. Version 0.2 took in RE-2's R-1 to R-8. This revision takes in RE-1's R-1 to R-6, RE-3's R-1 (its part here) and R-2 to R-5, RE-4's R-1 to R-6 and RE-5's R-1 to R-7, with the corrections each made to Section 2.3, together with Lahaina's R-7 and Katrina's R-7 from the coordination class.

10.6 Two Communications Design Basis Requirements Worded for the Electric Sector

CDB-2 and CDB-5 are read for answering and dispatch at Section 7.3 because their text names the electric sector. When the Military Action Communications Design Basis words them for every sector, Section 7.3 is revised to cite the new wording.

10.7 Which Dispatch Systems Can Be Pre-Authorized

Section 6.4 finds that advance discharge covers unit recommendations inside approved response plans, given a gatekeeper that holds a coverage floor, and does not yet cover machine-learned priority scores or determinant suggestions, because no proof now bounds their selections. Whether any dispatch system on the market has a gatekeeper of the kind *Mode of Discharge* §4.1 requires, separate from the recommender and unmodifiable by it, is not established and is the first thing a review of a center's systems should test. The sector qualification bar for the advance mode, which *Mode of Discharge* §11 leaves to sector annexes, is not set here.

10.8 Medical Direction and the Healthcare Boundary

Section 8.3 places the design basis of a medical dispatch action here and the clinical authority of the medical director at the healthcare boundary. On the author's decision of 6 October 2026 medical control divides at the clinical authority, and DBA-HPH_Sector_Framework §6.4 states the same line from its side. The framework's open item on the emergency-medical boundary (framework §9) may now record it as settled.

10.9 Conformance Owed by Other Instruments

Core Switching, Routing and Signaling may record at its next revision that a contracted 911 routing function which fails the contract test of Section 8.3 is on its side, and that the detection, alternate, invocation and notification obligations of Section 4.2 follow the call there (*Multistate 911 Outage* F-2, F-3). The framework adopts the recording of concentration at the sector level (Section 10.1). The vetting of reports at Section 4.3 answers *Response Forces and Stations* §10.10, and the false alert as a source of surge answers the Hawaii instrument's R-7, recorded in *Emergency Management and Coordination* §10.8. The derecho instrument returns to Access and Edge and to Core Switching, Routing and Signaling the monitoring that survives the loss of power, central-office backup power credited only when tested, and location-link diversity verified against the route (*Derecho 911 Outages* R-7); the call flood instrument returns to Access and Edge the origin obligation at the handset (*Call Flood* R-1); and its R-6, a human act at the point of origin as a sector principle, goes to the framework. Version 0.5 answers the finding *Care Delivery* §10.10 notes to this class from DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*), at Sections 4.6 and 8.3.

References

DBA-ESS Series and Parent Framework

Emergency Services Sector Framework (DBA-ESS_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

The Multistate 911 Outage of April 2014 (DBA-ESS-RE-2), *The False Ballistic Missile Alert in Hawaii* (DBA-ESS-RE-6), *Hurricane Harvey and the Houston Response of August 2017* (DBA-ESS-RE-10) and *The Route 91 Harvest Shooting of October 2017* (DBA-ESS-RE-11). Eclectic Technologies, 2026. Cited by the short titles *Multistate 911 Outage*, *False Missile Alert*, *Hurricane Harvey* and *Route 91 Harvest Shooting* and finding number.

The 911 Outages After the Derecho of June 2012 (DBA-ESS-RE-1), *The 911 Call Flood of October 2016* (DBA-ESS-RE-3), *The Ransomware Attack on Baltimore's Computer-Aided Dispatch, March 2018* (DBA-ESS-RE-4) and *The Ransomware Attack on the City of Dallas, May 2023* (DBA-ESS-RE-5); and, from the coordination class, *The Warning Failure in the Lahaina Wildfire* (DBA-ESS-RE-7). Eclectic Technologies, 2026. Cited by the short titles *Derecho 911 Outages*, *Call Flood*, *Baltimore Dispatch*, *Dallas Ransomware* and *Lahaina Wildfire* and finding or recommendation number.

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics.

Military Action Communications Design Basis (DBA-MA-CDB). Eclectic Technologies, 2026. CDB-1 through CDB-7 (§2) and the instantiation requirement (§3).

MA Applicability Framework (DBA-MA-AF). Eclectic Technologies, 2026. The four-filter applicability gate (§2, §3).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual form (§3.5).

Other Series

Access and Edge (DBA-COMM-AC-FC2). Eclectic Technologies, 2026. Critical-services elevation at the access boundary (§8.1).

Positioning, Navigation and Timing (DBA-COMM-PN-FC4). Eclectic Technologies, 2026. Timing receiver design basis for every sector (§7).

Communications Sector Framework (DBA-COMM_Sector_Framework). Eclectic Technologies, 2026. The critical-services overlay (§3.7).

Care Delivery (DBA-HPH-CD-FC1) and DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*). Eclectic Technologies, 2026. The receiving hospital's named path to command (§4.3) and the finding that a hospital on the area's radio is a user of it (§10.10).

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*) and CB-PAE (*Pre-Approved Action Envelope*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Events

Federal Communications Commission rules on 911 reliability, outage notification to 911 centers, enhanced 911 location and Next-Generation 911; National Emergency Number Association call-answering and Next-Generation 911 standards; National Fire Protection Association standard for emergency communications centers; TIA-102 (P25); Federal Emergency Management Agency alerting gateway (IPAWS).

Federal Communications Commission, *Impact of the June 2012 Derecho on Communications Networks and Services: Report and Recommendations*, January 2013, and the Commission's order of December 2013 on 911 reliability (FCC 13-158).

Federal Communications Commission, Public Safety and Homeland Security Bureau, *April 2014 Multistate 911 Outage: Cause and Impact — Report and Recommendations*, October 2014.

Public reporting of the telephony denial-of-service attack on 911 centers in Arizona and other states (October 2016), the ransomware attack on Baltimore's computer-aided dispatch (March 2018) and the ransomware attack on the City of Dallas (May 2023); the Independent Panel Reviewing the Impact of Hurricane Katrina on Communications Networks (2006).

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. The first facility class of the DBA-ESS series, typed physical on the author's decision of 6 October 2026. States the boundary with the communications sector where the call and the caller's location enter the 911 system, with the Emergency Services IP Network, its core services and agency-owned P25 systems in this class and FirstNet in DBA-COMM-AC-FC2, on the author's decisions of the same date. Defines operate-through duration and alternate-answering transfer time as the governing measures, on the author's decision of the same date, with the alternate-routing map as the third parameter. Proposes the derecho of June 2012, the multistate 911 outage of April 2014, the call flood of October 2016 and the ransomware attacks on Baltimore (March 2018) and Dallas (May 2023) as reference events, conferred by DBA-ESS_Sector_Framework v0.4 §9.1 as DBA-ESS-RE-1 to RE-5. Takes dispatch denied during a primary attack as the governing military-action scenario. Instantiates advance discharge under FD-BL-D1 for dispatch, on the author's decision of the same date, finding that it covers rule-bound unit recommendations and not yet machine-learned priority scores. Records at Section 10.3 what the boundary means for the communications series' critical-services overlay. Registers the DBA-MA-sense endurance requirement under Project 9 OI-16.
0.2 — DRAFT	October 2026	Tim Roxey	Conformed to DBA-ESS-RE-2 (*The Multistate 911 Outage of April 2014*), taking in its R-1 to R-8 and the author's contract test of 7 October 2026, and to the notes owed from DBA-ESS-RE-6, DBA-ESS-RE-10 and *Response Forces and Stations* v0.2 §10.10. Transfer time is bounded by recognition time (§3.2). The alternate-routing map records what invokes each transfer, what each alternate shares with its primary, and the population each upstream element serves; concentration is recorded there rather than made a parameter, on the author's decision of 7 October 2026 (§3.4). Detection is placed where failure is visible and rated by

Version	Date	Author	Description
			consequence, finite resources report their margin, alternates exclude shared design, and upstream transfer is automatic (§4.2); automatic failover is an above-line action authorized in advance (§6.4). A false alert and a regional disaster are named as sources of surge, overflow reaches outside the region, and reports are vetted before units are committed (§4.3). Silent exhaustion is an attack route (§5.4). Notification is triggered by detection (§7.3). The contract test replaces the contractor rule at §8.3, and the obligations follow the call. §10.1 and §10.5 conformed; §10.9 added. The running footer is brought to the house form.
0.3 — DRAFT	October 2026	Tim Roxey	Conformed to the class's remaining reference event instruments, DBA-ESS-RE-1 (*The 911 Outages After the Derecho of June 2012*), RE-3 (*The 911 Call Flood of October 2016*), RE-4 (*The Ransomware Attack on Baltimore's Computer-Aided Dispatch, March 2018*) and RE-5 (*The Ransomware Attack on the City of Dallas, May 2023*), and to the coordination class's RE-7 R-7 and RE-8 R-7, on the author's decisions of 7 October 2026. Section 2.3's accounts of all five events and of Katrina are corrected. Backup power is credited only when shown to run (§3.2, §4.1). The alternate-routing map holds the area's own evidence of the upstream elements, verifies diversity against the physical route including location links, and names overflow alternates by audience (§3.4). The operator's detection survives the loss of the power it watches, notification reaches a center at once, and the degraded states between silence and service are planned (§4.2). A generated flood is recognized by behavior; the time to remove its sources is in its design duration; the origin obligation and a disaster line are named (§4.3, §5.4, §8.2). The manual mode covers what is lost with dispatch, keeps the record, and is shown by measurement recorded under CDB-7, not by a new parameter; a planning duration for manual operation after a network-wide attack is stated with its value open (§4.4, §7.3, §10.1). Separation includes identity and administration, a change to protection is a governed act at one center, and dispatch data is protected against disclosure (§4.7). The decision to take dispatch offline is prepared (§5.5). The Next-Generation 911 confirmation is recorded and the contract binds the provider to supply evidence (§8.3). The federal reliability rule is stated as revised in June 2026 (§8.4). §10.3, §10.5, §10.8 and

Version	Date	Author	Description
			§10.9 conformed; OI-24 recorded closed. Issued under framework v0.7. Revision-history entries are left as records.
0.4 — DRAFT	October 2026	Tim Roxey	Takes in the findings of the series' first use cases, on the author's decisions of 11 October 2026, applying Emergency Services Sector Framework v0.10 §3.5 and §4.2. From DBA-ESS-UC-1 (The Compound Failure the Sector Must Operate Through): the center's operate-through duration as one link in the sector's chain and shared stocks counted once (Section 3.2), and the heartbeat (Section 7.2). From DBA-ESS-UC-2 (Dispatch Corrupted, Not Encrypted, During a Primary Attack): a unit-originated, authenticated check against the dispatch system and a sufficient indication for taking it offline (Section 5.5); the envelope compared with an approved copy, and the return to automation authorized by its approvers (Section 6.4); restored systems checked before use (Section 4.7); manual operation stated in staff (Section 4.4); and a third declared residual, the interval from corruption to recognition (Section 5.7). From DBA-ESS-UC-3 (A Second Strike Timed to the Response): the envelope frozen during a deliberate attack, and a named holder for dispatching to reported attacks (Section 6.4). Revision-history entries are left as records.
0.5 — DRAFT	October 2026	Tim Roxey	Takes in the finding DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*) notes to the Emergency Services series, recorded at *Care Delivery* §10.10, on the author's decision of 12 October 2026. Section 4.6 counts among the users the radio system serves under denial each receiving hospital whose named path to command is a terminal on it, requires the system to state the site or channel by which each such hospital remains reachable when sites operate on their own and traffic moves to mutual-aid channels, and does not credit a hospital reachable only through the controller or through backhaul the same event can cut; the terminal, its power and its operators stay the hospital's. Section 8.3 records the joint with Healthcare, and Section 10.9 the answer. References add *Care Delivery* and the use case. The cover lineage stays as issued. No change to the class's scope, chains, measures or placements; earlier entries are left as a record.